



Figure 2: 32-node cluster at TCE data centre

available actions will be displayed. The administrator can define new roles and flavours, administer the cloud and upload new images, which are stored in a MySQL database, and are managed by the Glance component. We have scripted image uploading. Users can choose from images as per their requirements, then allocate resources based on predefined flavours, and launch the images. The Nova Scheduler will then assign this instance to one of the nodes in the cloud; Nova Compute will then provide the computing power, and Nova Volume will provide storage services. Nova Network handles communication between nodes and images; it creates a bridge between the private network and the virtual network. It also handles Network Address Translation (NAT). VMs are assigned IP addresses via DHCP.

The Nova API provides metadata and additional features to images. Users can access instances through a VNC console (Figure 3) via the dashboard in most major browsers, including Firefox, Google Chrome and Internet Explorer. This is provided by the VNC Proxy service, which is responsible for displaying the requested virtual machine. QEMU is the hypervisor, and it supports hardware virtualisation with KVM. Once users log into their VMs, they can perform their tasks with the allocated resources. These instances persist until they are terminated manually. Users can also store snapshots at any time, for later restoration.



Figure 3: Ubuntu VM accessed through the VNC console

## Scope for development

Since OpenStack is highly scalable, based on the requirements of users, additional resources can be added to the cloud at any time. Also, because it is open source, students have an opportunity to contribute to its development by writing additional modules. Research can be carried out on optimising different components such as the scheduler, network controller, storage controller and compute controller.

We are planning to provide additional storage by mounting NFS exports from other nodes. We are also planning to create and upload images with additional software and services for the benefit of students and faculty alike. These include software for CAD and circuit design. Newer releases of OpenStack support attaching new volumes to instances, dynamically, at runtime. Once these releases are stable, they may be adopted at TCE.

We hope that in the years to come, the TCE Cloud will be the preferred way for researchers and students to carry out their projects. The students involved in setting it up gained a lot of experience, and got to enjoy cloud computing first-hand. **END** 

## References

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